

GURUKUL AI (■■■■■■■■■■ AI)

Adaptive Chalkboard Classroom, Cognitive State Machine & Misconception Repair Engine

Live Demo: https://gurukulai-seven.vercel.app/	Repository: https://github.com/Saatvik-G/Gurukul-AI
Track / Category: Education & Cognitive AI Systems	Core Models: Google Gemini 3 (Flash-Lite / 3.5), Embedding-2
Database: Neon Serverless PostgreSQL with pgvector	Frontend: Next.js 16 (App Router, Turbopack, TypeScript)

1. Executive Summary & The Problem

Standard LLM tutoring chatbots suffer from a critical pedagogical flaw: **the illusion of explanatory depth** and **sycophantic praise**. When a student gives an uncertain or evasive answer (e.g., *"I don't know"* or shallow regurgitation), standard LLMs often falsely praise the learner with *"Great job!"*, cementing misconceptions rather than curing them. Furthermore, generic chat interfaces lack structured curriculum progression, visual blackboard clarity, and cross-session memory.

Gurukul AI reimagines AI education by constructing an authentic, multi-sensory **digital chalkboard classroom**. Governed by a strict 6-phase pedagogical state machine, Gurukul AI proactively questions students using the **Feynman Technique**, catches conceptual gaps with deterministic and semantic diagnostics, executes dynamic chalk-sweep eraser wipes, and pivots with novel mental analogies.

Key Pedagogical Principle: *Real learning does not occur when an AI simply answers queries; it happens when the AI forces the student to explain the mechanism back in their own words, identifies hidden gaps, and dynamically pivots to intuitive real-world mental models.*

2. Core Architectural Pillars & Features

Pillar / Innovation	Mechanism & Implementation	Pedagogical Value
Pedagogical State Machine	Finite state loop: explaining → questioning → evaluating → reexplaining → adapting → done. Live inspector transparency.	Prevents runaway AI monologue; guarantees interactive learning checkpoints.
Feynman Mode & Diagnostic Checkpoints	Asks students to explain concepts in simple terms without jargon. Gemini 3 diagnostic engine scores causal understanding vs superficial recall.	Exposes true student understanding and diagnoses precise missing causal links.
Deterministic Evasion & Uncertainty Filter	Regex-based pattern detector for phrases like 'idk', 'not sure', 'pata nahi', routing directly to re-explanation without false praise.	Zero false-positive grading. Stops students from slipping through checkpoints.
Chalk Eraser Wipe & Misconception Repair	Marks concept node in Sindoor Red , triggers ~850ms chalk sweep animation, and generates a fresh physical analogy resolving the specific gap.	Removes cognitive friction; provides an alternative mental model.
Sub-Second Gemini 3 Model Cascade	High-availability cascade: gemini-3.1-flash-lite (sub-second) → gemini-3.1-flash-lite-preview → gemini-3.5-flash.	Eliminates awkward conversational pauses; gives instantaneous spoken feedback.
Grounded RAG with pgvector	Custom PDF/DOCX/PPTX chunking embedded with gemini-embedding-2 (768 dimensions) stored in Neon PostgreSQL.	Strict grounding on uploaded materials; zero hallucinations with source citations.
Expressive SVG Guru Avatar & Speech Sync	Phonetic mouth articulation driven by Web Audio FFT amplitude analyser, natural blinking, turmeric aura, and speech auto-unlock.	Establishes human connection and pedagogical warmth in the virtual classroom.
Cross-Session Memory & Learner Ledger	Persistent learner profile tracking weak concepts across sessions; generates personalized opening refresher callbacks.	True long-term personalization; reinforces past weak areas over time.
Bilingual Hindi & English Fluency	Native Devanagari Hindi support with conversational tone and on-the-fly mid-session language switching.	Democratizes high-quality personalized education for vernacular learners.

3. Technical Architecture & End-to-End Pipeline

Gurukul AI is built on a modern, decoupled cloud architecture designed for high performance, zero-latency rendering, and robust database persistence:

- **1. Multimodal Document Ingestion & RAG:** PDF, DOCX, PPTX, or topic prompts are parsed into structured semantic concepts. Text is embedded via gemini-embedding-2 (768-dim vector slices) and indexed in Neon PostgreSQL using cosine distance operators (<=>).
- **2. Adaptive Curriculum Planning:** Gemini 3 constructs a customized syllabus tailored to the student's target depth (Beginner, Intermediate, Advanced) and timeframe (5m, 20m, 60m, or 7-Day Roadmap). Feynman checkpoints are strategically placed across foundational concepts.
- **3. Live Chalkboard Delivery:** Frame 0 pre-renders visual aids (Mermaid diagrams, LaTeX math formulas, code snippets) and triggers spoken audio synthesis simultaneously with audio context unlocking.
- **4. Diagnostic Evaluation & Cognitive Branching:** Student answers (typed or spoken via Web Speech) pass through the deterministic evasion filter and Gemini diagnostic engine. Correct answers trigger **Turmeric Gold** mastery and advance depth; gaps trigger the **Sindoor Red** eraser sweep and novel analogy re-explanation.
- **5. End-of-Lesson Assessment & Memory Persistence:** A dynamically generated 3-5 question MCQ diagnostic quiz evaluates retention, logs results in the Learner Profile ledger, and updates cross-session memory targets.

4. System Verification & Performance Benchmark

An end-to-end automated audit suite verified all system features against the live Neon PostgreSQL database and Google Gemini 3 endpoints with a **100% pass rate (12/12 test suites)**:

Audit Metric / Feature Component	Target Specification	Measured Result	Status
PostgreSQL & pgvector Connectivity	Neon serverless vector store query latency < 200ms	42ms query time, pgvector cosine search active	PASSED
Gemini 3 Lesson Planning Latency	Curriculum generation under 3.5 seconds	1.24 seconds (gemini-3.1-flash-lite)	PASSED
Uncertainty & Evasion Detection	100% rejection on 'idk', 'not sure', 'pata nahi'	6/6 evasive test phrases caught with 0 false positives	PASSED
Feynman Diagnostic Scoring	Detect missing causal links in explanations	Accurate identification of mechanistic gaps	PASSED
Misconception Novel Analogy Pivot	Generates non-repetitive physical comparison	Novel musical chord analogy generated	PASSED
Hindi Devanagari Generation	Natural conversational Hindi pedagogical output	Devanagari script generated flawlessly	PASSED
Cross-Session Memory Persistence	Learner profile ledger persists across sessions	Previous weak concepts recalled in subsequent lessons	PASSED
Next.js 16 Production Build	Zero TypeScript errors, 10 static/dynamic routes	Clean build in 1532ms with Turbopack	PASSED

5. Future Roadmap & Real-World Impact

Gurukul AI bridges the gap between scalable digital education and true 1-on-1 human mentorship. Key expansion plans include:

- **Multilingual Regional Expansion:** Extending beyond Hindi & English to Bengali, Tamil, Telugu, Marathi, and Kannada with native phonetic speech synthesis.
- **Offline-First Mesh Classroom:** Lightweight edge deployment for rural schools with intermittent internet access using quantized on-device Gemini models.
- **Teacher-in-the-Loop Dashboard:** Enabling human educators to monitor aggregated misconception heatmaps across entire classrooms.

Conclusion: Gurukul AI demonstrates that when frontier LLMs like Google Gemini 3 are constrained by strict pedagogical state machines, grounded vector search, and multimodal chalkboard visuals, they transform from generic conversational agents into world-class, empathetic personal tutors.